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CS-499

Prof Ogoh

Enhancement 2 Narrative: Algorithms and Data Structures

The artifact I selected for the second enhancement is the Courser Advisor program that I had made for CS-300. The program was initially built to load data about courses and allow the users to view each course's information with prerequisite and course information included. The application is built on a menu driven concept that the user interacts with the search and displays the relevant information in a structured format.

I chose this artifact as it allows me to demonstrate the use of algorithms and data structures to provide an organized method for efficient data retrieval. I plan to implement a hash table that could access course records through a key style lookup instead of iterations of all entries. This will bolster the applications' performance and show a much deeper understanding of how data structures impact the efficiency of an algorithm. Additionally, this shows how improving the organization of the code into clearer components for loading, storage and retrieval will allow the program to be easier for others to understand while reinforcing the relationship between design and software structure.

The enhancements I plan to make for this artifact support the course outcomes in relation to the design and evaluation of computing solutions through algorithmic principles and appropriate data structures. The hash table for example will showcase my ability to apply efficient data structures to solve problems within a system and improve performance as well. These changes are meant to help support the outcomes that relate to using the appropriate

techniques and tools. The sequential search approach via a hash-based lookup shows how I can make a conscious design decision that impacts time complexity and scalability.

One challenge I encountered during this process was when I needed to restructure the logic to integrate the new data structure without breaking the system's functionality. Ensuring that the courses data was loaded appropriately and retrieved required several tests and adjustments to the flow of the program. This process reinforced how important selecting the appropriate data structure early in the development and its impact on both performance and maintainability.